SIMATS ENGINEERING

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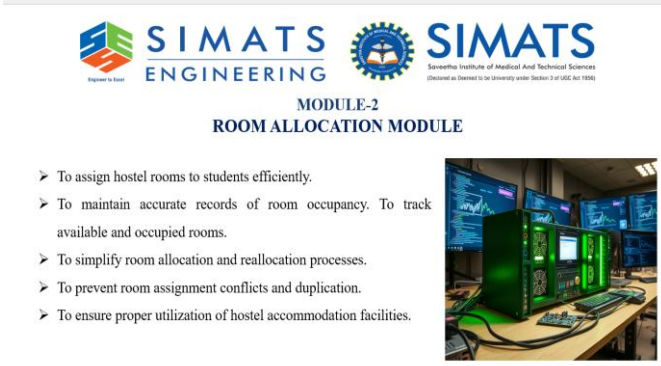
Course Code: DSA-0126 **Slot:** A

Course Name : Object Oriented Programming using C++ for Modular Programming

Course Faculty : Dr.R.KESAVAN

Project Title: HOSTEL MANAGEMENT SYSTEM USING LINKEDIN LIST IN C++

Module Photographs:





Project Description:

A Hostel Management System is a software application designed to automate and simplify hostel administration activities such as student record management, room allocation, and fee tracking. Traditional hostel management methods rely heavily on manual record keeping, which is time-consuming, prone to errors, and difficult to maintain as the number of students increases. To overcome these challenges, a computerized system is developed using C++ programming, Object-Oriented Programming (OOP) concepts, and Linked Lists. In this project, student information is stored dynamically using a linked list data structure, allowing efficient memory utilization.The system provides functionalities such as adding new student records, searching existing records, updating information, and deleting outdated entries. A room allocation module is included to manage room assignments, track occupancy status, and prevent allocation conflicts. Additionally, a fee management module helps administrators maintain payment records and monitor pending fees accurately. The use of Object-Oriented Programming improves modularity, code reusability, and maintainability of the system. Linked Lists enable dynamic storage without requiring fixed memory allocation, making the application flexible and scalable.The system reduces manual effort, improves data accuracy, enhances administrative efficiency,and hostel information. This project demonstrates the practical application of data structures and OOP concepts in solving real-world management problems effectively.

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